SHAJEE RAZA

AI Engineer | LLM, Multi-Agent Systems & RAG

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years building LLM applications, multi-agent systems, and production RAG pipelines, holding an M.Sc. in Big Data and Machine Learning from ITMO University (Russian Open Doors Scholarship, 2024). Delivered a hallucination detection algorithm that raised SWE-bench Lite correctness from 0.58 to 0.70, a 20.7% relative gain, and engineered fault-tolerant Apache Spark pipelines over 400,000+ record datasets. Works end to end across LangGraph, LangChain, PyTorch, FastAPI, Spark, Docker, and GitLab CI/CD.

CORE TECHNICAL SKILLS

AI & Machine Learning: Python, PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, Deep Learning, NLP, Deep Autoencoders, Computer Vision, Document AI / OCR, Model Evaluation, Feature Engineering

LLMs & Multi-Agent Frameworks: LangGraph, LangChain, LangSmith, OpenHands SDK, RAG, Auto-RAG, Multi-Agent Systems, Agentic Workflows, Hallucination Detection & Mitigation, LLM Evaluation, SWE-bench Lite, Prompt Engineering, Multimodal LLMs (MM-LLMs), Model Alignment, Pydantic, Vector Databases (ChromaDB, Pinecone), Embeddings

Distributed Systems & Big Data: Apache Spark, Spark MLlib, Hadoop, Parquet, Distributed Data Pipelines, Fault-Tolerant Architectures, Graceful Shutdown, Pandas, NumPy

Backend & Tools: FastAPI, REST APIs, Streamlit, MySQL, SQL, DBMS, Docker, Linux, Git, GitLab CI/CD, Jupyter, C, C++

PROFESSIONAL EXPERIENCE

Research Intern, Industrial AI Lab | ITMO University | St. Petersburg, Russia | Oct 2024 to Jun 2026

- **Cut hallucinated code output** by designing a 5-dimension scoring rubric (Correctness, Edge Cases, Security, Code Quality, Relevance) for a Reviewer Agent inside a LangGraph multi-agent code-generation system, lifting SWE-bench Lite correctness from 0.58 single-shot to 0.62 reviewed to 0.70 under a confidence-gated refine loop, a 20.7% relative gain.
- **Generalized hallucination detection to autonomous tool-use agents** by engineering a Hallucination Bridge that audits structured OpenHands ActionEvent and ObservationEvent traces, extending coverage beyond code generation without modifying the base agent.
- **Benchmarked 4 document chunking strategies** (fixed-size, sentence-based, semantic, hybrid) for an Auto-RAG question-answering system and shipped a hybrid sentence-boundary plus semantic-threshold method that improved retrieval coherence at no measurable latency cost.
- **Built a reproducible LLM evaluation stack** on LangGraph, LangChain, ChromaDB with all-MiniLM-L6-v2 embeddings, and Pydantic schema validation, enabling repeatable scoring runs across the SWE-bench Lite benchmark suite.

Backend Python Developer & Data Analyst | ITMO University, National Centre of Cognitive Research | St. Petersburg, Russia | Feb 2025 to May 2025

- **Eliminated data loss on pipeline termination** by architecting a graceful shutdown mechanism for Apache Spark applications covering normal exit, exception, and OS signal paths, guaranteeing safe resource cleanup on every route.
- **Reduced shutdown boilerplate by approximately 60%** across a multi-developer codebase by implementing a reusable graceful_exiter context manager and GracefulExiter class adopted as the project-wide standard.
- **Refactored pipeline operator classes** onto a consistent initialise, finalise, and exit interface, shipping every change through Git feature branching and GitLab CI/CD validation.

AI Workflow Automation Specialist | Headstart School Pvt. Ltd. | Islamabad, Pakistan | Oct 2023 to Dec 2023

- **Designed and deployed AI-powered automation frameworks** in Python to streamline administrative processes across admissions, HR, and procurement.
- **Developed intelligent co-pilot tools** that reduced manual effort and improved operational efficiency across departments.
- **Audited departmental workflows** to identify recurring inefficiencies and implemented AI-driven solutions tailored to each process.

ML / AI Intern | Joyn Digital | Islamabad, Pakistan | Jul 2022 to Sep 2022

- **Performed data preprocessing and model development** for AI projects, working hands-on with Python, TensorFlow, and PyTorch.
- **Delivered practical AI solutions to real-world project challenges**, covering the path from raw data cleaning through model training and evaluation.

FEATURED PROJECTS

Multi-Agent Study & Coding Assistant | LangChain, LangGraph, ChromaDB, FastAPI, Streamlit | Oct 2025 to Jan 2026

- **Architected a 4-agent routing system** dispatching user queries to specialised research, coding, writing, and study agents, backed by persistent ChromaDB vector memory, live web search, and sandboxed code execution, served through a FastAPI backend and Streamlit interface. github.com/shajeeraza12/Multi-Agent-Study-Coding-Assistant

TMDB Big Data Pipeline | Apache Spark, Spark MLlib, Parquet | ITMO University, M.Sc. Programme

- **Processed 400,000+ movie records** through a distributed Spark pipeline handling nested JSON normalization, null treatment, and Parquet storage, then trained a Random Forest genre classifier in Spark MLlib to forecast genre demand.

PIEAS FAQ Assistant & Document AI / OCR Pipeline | NLP, Deep Autoencoders, OCR | Sep 2022 to Jun 2023

- **Engineered a deep-autoencoder conversational agent** for student admissions and campus queries as the B.Sc. final year project, and separately delivered a paid freelance OCR pipeline mapping poorly scanned PDFs into structured CSV columns.

RESEARCH & SELECTED WORKS

- **Lightweight Hallucination Detection for the Reviewer Agent in a Code-Generation Multi-Agent System.** M.Sc. thesis, ITMO University, Sep 2025 to Jun 2026. Supervisor: Dr. Nasonov Denis Aleksandrovich. LangGraph reviewer agent with a 5-dimension rubric, evaluated on SWE-bench Lite.
- **Automated Document Chunking for Auto-RAG Question Answering Systems.** ITMO Industrial AI Lab, Feb 2025 to Jun 2025. Benchmarked 4 strategies and proposed a hybrid method for real-time QA pipelines.
- **Evolution from NLP to Multimodal Large Language Models: An Analytical Review.** ITMO Industrial AI Lab, Sep 2024 to Jan 2025. Surveyed BERT and GPT architectures and identified 3 open gaps in modality alignment, cross-modal scalability, and interpretability.

EDUCATION

M.Sc. in Big Data and Machine Learning (EQF Level 7) | ITMO University, St. Petersburg, Russia | Sep 2024 to Jun 2026

Thesis: A Lightweight Hallucination Detection Algorithm for the Reviewer Agent in a Code-Generation Multi-Agent System. Supervisor: Dr. Nasonov Denis Aleksandrovich. Coursework: Reinforcement Learning, Advanced Machine Learning, Neural Networks for Deep Learning, Advanced NLP.

B.Sc. in Computer and Information Sciences (EQF Level 6) | Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad, Pakistan | Sep 2019 to Jun 2023

Final year project: deep-autoencoder conversational agent for institutional FAQ handling. Coursework: Computer Vision, Computational Intelligence.

HONOURS

- **Russian Open Doors Scholarship, 2024 (World Rank 30).** Competitive international merit scholarship awarded in Big Data and Machine Learning, funding the full M.Sc. programme at ITMO University.